

Quantitative Cross-Cultural Analysis of Romantic Preferences

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I. ABSTRACT

This study develops and evaluates machine learning models to predict self-reported infidelity using a large cross-cultural dataset of 117,293 participants from 175 countries. Using logistic regression and XGBoost, we model infidelity as a binary classification task based on demographic, behavioral, and attitudinal variables. Model performance is assessed using calibration-focused metrics, including the Brier score, alongside accuracy and F1 score. Results indicate that relationship-specific behavioral and moral factors particularly sociosexual behavior and attitudes toward infidelity are the strongest predictors, while broader demographic traits contribute minimally. These findings highlight consistent cross-cultural patterns while underscoring ethical considerations in modeling sensitive human behavior.

II. INTRODUCTION

The aim of this project is to develop two machine learning models that predict whether an individual has cheated on their partner based on a set of personal characteristics. This problem is a binary classification task, in which the outcome is whether an individual's behavior aligns with infidelity or not. We chose this problem because our data set provides a rich mix of demographic, behavioral, and attitudinal variables, offering an interesting opportunity to explore how machine learning can be used to analyze aspects of human behavior within romantic relationships. Given the complex nature of infidelity, which is influenced by various personal, relational, and moral factors, machine learning can be used to identify patterns in data that can otherwise be difficult to discern. The models developed in this project use characteristics such as demographics, moral beliefs, and behavioral traits to generate predictions. The primary goal of this project is to create an accurate and well-calibrated model that can predict if an individual will cheat. This model is discussed in the context of applications such as those under the Korean Civil Act, where estimates of infidelity could influence legal proceedings. At the same time, this project also acknowledges and addresses potential biases and ethical implications inherent in making predictions about sensitive human behavior.

III. DATA

A. Data Description

The data used in this project come from a large cross-cultural survey conducted by an international team of researchers led by Dr. Kowal, a psychology researcher from the

University of Wrocław, containing responses from 117,293 participants from 175 countries. The survey collected extensive information on participants' demographics, relationship behaviors, sociosexual attitudes, mate preferences, relationship satisfaction, jealousy, gender attitudes, collectivist values, and self-reported infidelity behaviors (Kowal et. al, 2025). This broad range of psychological, attitudinal, and demographic measures can serve as meaningful indicators for predicting self-reported infidelity.

The outcome variable for this study is whether a participant reported engaging in infidelity. This variable is encoded as a binary indicator, where 0 represents no reported cheating and 1 represents reported cheating. The outcome is taken directly from the survey item asking participants if they have ever cheated on a partner, in which 38% of respondents reported infidelity while the rest said they had not cheated. A binary classification is appropriate for this task because the goal of the analysis is to predict how likely an individual is to cheat based on a set of given characteristics.

To identify which variables were suitable predictors for the modeling task, ANOVA and Chi-square tests were used to assess the relationships between individual features and the binary outcome. The variables selected to be used in the models were those showing significant associations with reported infidelity, as well as those that were practically relevant. The final set of characteristics for the models included sociosexual behavior, sexual attitudes, level of desire, level of jealousy, moral views of infidelity, indicators of relationship quality, age, religion, employment status, gender attitudes, political identity and class identity. These features capture meaningful behavioral, attitudinal, and demographic dimensions that can influence the likelihood of reporting infidelity, thus serving as appropriate predictors in this analysis. These features are also unlikely to cause data leakage compared to metrics such as type of infidelity, which is an outcome of cheating status.

B. Data Preprocessing

Categorical variables including age, religious affiliation, employment status, and political identity were transformed using one-hot encoding to convert each category into a set of binary indicator variables. Continuous variables such as sociosexuality scores, jealousy measures, moral rating, and relationship quality indicators were retained in their original numeric form. This preprocessing approach preserved the interpretability of the features.

The data was also partitioned into training and testing sets using an 80/20 split with stratified sampling based on cheating status to preserve class proportions. A stratified split was chosen due to the class imbalance in the cheating status tar-

get variable. By stratifying it, it minimized the chances of the model receiving a non-representative training dataset.

IV. PREDICTIVE MODELING

A. Logistic Regression

Logistic regression was selected as a potential model due to its interpretability, computation efficiency, and relatively fast training time compared to more complex models. Linear regression was not selected because the outcome variable is binary. Linear models are not designed to model probabilities, whereas logistic regression is specifically suited for binary classification. In terms of interpretability, the resulting logistic coefficients tell us directly how much each feature matters and in what direction. Since figuring out what features cause cheating can lead to better insights about romantic relationships, being able to explain why the model makes certain predictions is just as important as getting the correct prediction.

Model optimization was performed using 5-fold stratified cross-validations in combination with GridSearchCV to systematically tune hyperparameters. The grid search evaluated multiple solver algorithms ('liblinear', 'lbfgs', 'saga') as well as maximum iteration values (100, 200, 500). The optimal configuration was identified as the 'liblinear' solver with a maximum of 200 iterations, yielding a cross-validation F1 score of 0.6914. The 'liblinear' solver is particularly well-suited for binary classification problems and supports both L1 and L2 regularization, which helps control model complexity, reduce overfitting, and preserve interpretability. Additionally, although the F1 score was employed for model selection, calibration, quantified by the Brier Score, was also examined as a primary evaluation metric to ensure the reliability of probability estimates.

B. XGBoost

The second model chosen was the XGBoost, which is a model that sequentially runs weaker decision trees, while adjusting for errors to make a better overall model. This was chosen as it outperforms random forest, is quicker and more efficient than regular gradient boosting, and is inherently robust in preventing overfitting. This model also works well in datasets with both categorical and numeric features, and allows for feature importance analysis for interpretability. However, it is more complex and computationally expensive compared to many other simpler models, and requires proper parameter tuning.

A logistic loss function was used as the loss function as it is the standard for XGBoost and is meant for binary outcomes. Using the grid search cross validation on the negative of the brier score, because the goal is for the best calibration, the best max depth of the trees was found to be 4 and the number of estimators to be 300. As for the col sample by tree, the subset ratio of random features for each tree to help prevent correlations between trees, the best was 1.0, and the learning rate, or the weight of each new feature from the last tree, was

0.05, and the subsample, i.e. the ratio of the training data random sampling to prevent overfitting, was 0.7.

V. RESULTS

To evaluate the models, we once again emphasized the brier score as our main metric because calibration was the goal, as it better reflects real world likelihoods. The weighted F1 score was also chosen as it balances precision and recall in proportion to the data representation, which is useful for unbalanced datasets like ours. Accuracy, precision, and recall were all also chosen as a baseline for model performance in comparison to the brier and F1. We also considered energy and time usage since efficiency in model creation was an important factor to consider long term for any model.

A. Logistic Regression

Running this model required minimal computational resources, with training consuming approximately 0.000036 kWh of energy and producing effectively 0.0 kg of CO2 emissions over a runtime of approximately 2.48 seconds. Prediction energy usage was also negligible.

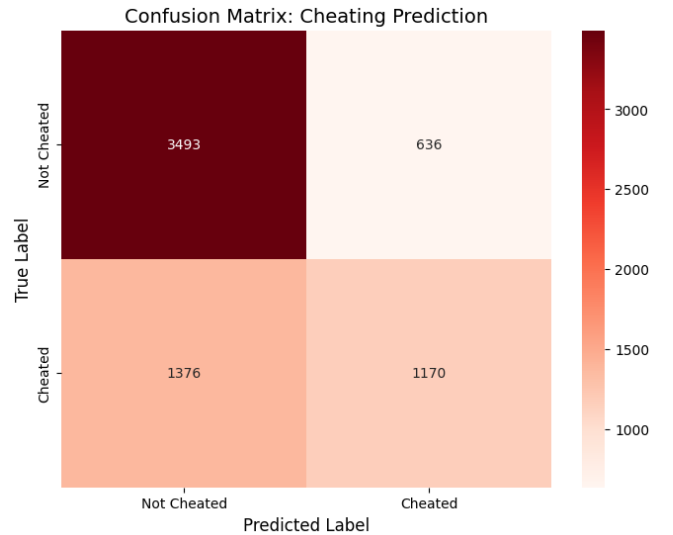


FIG. 1. Logistic Regression Confusion Matrix

When evaluating the confusion matrix, the model performed better at correctly identifying individuals who did not cheat than those who did (FIG 1). While false positives remained relatively low, the number of false negatives was notably higher, indicating that the model struggled more with detecting cheating behavior and was more likely to incorrectly classify a cheating individual as a non-cheater.

The logistic regression model achieved a Brier score of 0.1886, indicating that its predicted probabilities are reasonably well-calibrated, with values moderately close to the ideal score of zero. The model attained an accuracy of 0.7094, precision of 0.7028, recall of 0.7094. These values combine to produce an F1 score of 0.6972, reflecting a balanced trade-off

between precision and recall. In addition to our other metrics, the model was evaluated using the logarithmic loss (log-loss), which measures the quality of the predicted probabilities by penalizing confident but incorrect predictions. Log-loss is an appropriate loss function for probabilistic binary classifiers because it directly assesses how well predicted probabilities align with the true class labels. The logistic regression model achieved a log-loss of 0.5612, showing moderate probabilistic accuracy.

Collectively, these metrics indicate that while the model demonstrates stable and balanced performance, there remains substantial room for improvement, particularly in enhancing the detection of cheating cases (sensitivity) without introducing additional false positives. This suggests that, although logistic regression provides a strong interpretable baseline, more flexible models may be needed to capture the full complexity of infidelity-related behavior.

The top predictors were identified based on the absolute magnitude of the regression coefficients and are shown below:

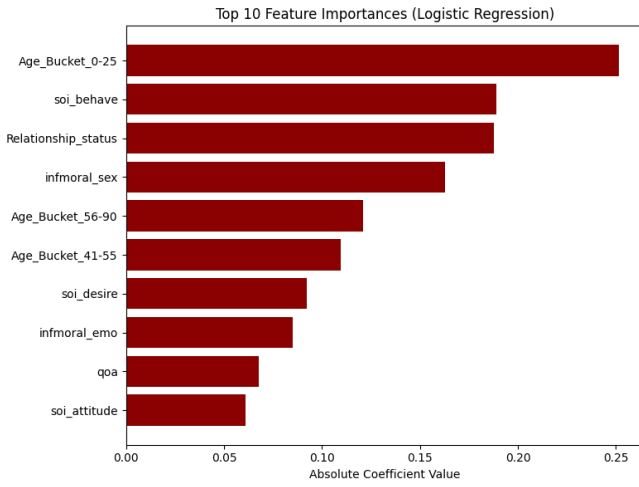


FIG. 2. Top 10 Feature Importance (Logistic Regression)

Among the top ten features, Age_Bucket_0–25 is the most influential, accounting for about 24% of the total absolute coefficient weight (FIG 2). The next most important feature is sociosexual behavior (soi_behave) at around 21%, followed by moral views toward sexual infidelity (infmoral_sex) at around 16% and relationship status at around 14.5%. Together, these four features make up over half (75%) of the total importance in the plot. The remaining variables each contribute a smaller share: Age_Bucket_56–90, Age_Bucket_41–55, perceived alternatives (qoa), moral views toward emotional infidelity (infmoral_emo), sexual desire (soi_desire), and Age_Bucket_26–40. This distribution shows that the model’s predictions are driven primarily by age, sociosexual behavior, moral evaluations of infidelity, and relationship status, with other factors playing smaller roles.

B. XGBoost

Running the XGBoost took 17 and a half minutes and consumed 0.015509 kWh of energy and emitted 0.004151 kg of

CO2 to train the model. The prediction energy usage was negligible. Its mean score from a stratified 5 fold cross validation was -0.1973 with a low variance, which shows the model struggled with unseen data.

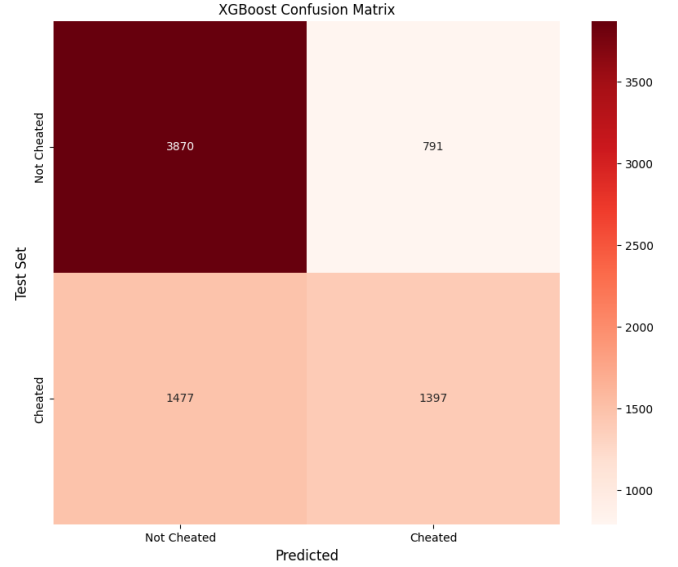


FIG. 3. XGBoost Confusion Matrix

From the XGBoost confusion matrix, the model was relatively performed well at predicting those who did not cheat with 86% accuracy (FIG 3). However, it struggled heavily with those who did cheat. Though it did correctly predict over half the time, it only managed to do so at 53%, making it only slightly better than a coin flip. Indicative of this was the logarithmic loss was 0.533, which shows that the XGBoost only correctly identified the infidelity of a person 58.7% of the time. However, the mean brier loss was 0.178, which is close to zero, showing that the model is calibrated quite well. It had an accuracy score of 0.733, precision score of 0.728, a recall score of 0.733, and, finally, a F1 score was 0.724. Though these values are all somewhat close to 1, there is still much room for improvement in the model, especially in its predictions for cheating individuals.

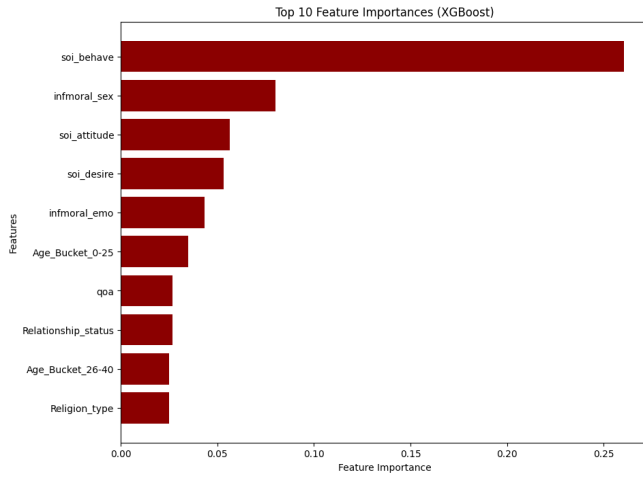


FIG. 4. Top 10 Feature Importance (XGBoost)

From the graph above, the most important feature was *soi_behave*, which measures a participant's number of casual partners (FIG 4). This feature accounted 31.4% of the results. Meanwhile, other features accounted for much less with the second most important feature being *infmoral_sex*. This variable measures a participant's attitude to casual sex by how comfortable they would be in having casual sex. This feature accounted for 9.9% while all other features only accounted for 5% or less of the results. The least important feature was whether the person would be jealous if their partner had an emotional relationship with someone else, at 1.8%.

VI. EVALUATION

A. Logistic Regression

Because logistic regression is an inherently interpretable model with directly observable feature coefficients and effect directions, we instead present a table of coefficients and corresponding odds ratios, which provide direct, quantitative interpretations of how each predictor changes the likelihood of infidelity. Odds ratios are particularly well-suited for this context because they translate model effects into meaningful changes in risk, making them more transparent and informative than methods such as SHAP values or partial dependence plots, which are primarily designed for black-box models.

Feature	Coefficient	Odds Ratio (OR)
Age_Bucket_0-25	-0.223687	0.799565
soi_behave	0.203185	1.225299
infmoral_sex	-0.161362	0.850984
Relationship_status	-0.146546	0.863686
Age_Bucket_56-90	0.114954	1.121821
Age_Bucket_41-55	0.103022	1.108515
qoa	0.098780	1.103823
infmoral_emo	-0.088276	0.915509
soi_desire	0.073847	1.076642
Age_Bucket_26-40	-0.070766	0.931680
soi_attitude	0.058108	1.059829
cot_class	-0.048988	0.952193
cot_politics	-0.041597	0.959256
Religion_type	-0.040829	0.959993
cot_educ	0.039736	1.040536
self_physatt	0.028487	1.028896
Individualism_col_3	-0.027700	0.972680
Individualism_col_2	0.024630	1.024936
relmob	0.023341	1.023615
Employment	-0.022476	0.977775
self_kind	-0.020354	0.979852
Gender_Equality_3	-0.016388	0.983746
jealemo	0.016177	1.016309
Gender_Equality_1	-0.009959	0.990091
Gender_Equality_2	0.009741	1.009788
jealsex	0.009254	1.009297
Individualism_col_1	-0.005811	0.994206
Individualism_col_4	0.004835	1.004847

TABLE I. Logistic Regression Coefficients and Odds Ratios

Based on the table, the results reveal that sociosexual behavior (*soi_behave*) is one of the strongest predictors of self-reported infidelity (OR = 1.23), indicating that each additional casual sexual partner is associated with 22.5% increase in the odds of cheating, holding all other variables constant (TABLE 1). In contrast, individuals aged 0-25 show lower odds of infidelity (OR = 0.80). Moral views toward infidelity are also influential: perceiving sexual infidelity as more morally wrong reduces the odds of cheating (OR = 0.85), as does stronger opposition to emotional infidelity (OR = 0.92). Relationship status further acts as a protective factor (OR = 0.86), reflecting the stabilizing role of commitment. Age effects vary across life stages, with individuals aged 41-55 and 56-90 showing modestly higher odds of cheating, while those aged 26-40 have slightly lower odds. Perceived relationship alternatives (*qoa*), sexual desire (*soi_desire*), and comfort with casual sex (*soi_attitude*) all increase cheating risk, though less strongly than sociosexual behavior. In contrast, broader demographic and ideological variables (including education, class, political identity, religion, employment, individualism, and gender-equality attitudes) have odds ratios near one, indicating minimal practical influence once relationship-specific and attitudinal factors are considered.

Overall, the pattern of strong versus weak predictors supports the theoretical insight that infidelity is primarily driven by relationship-specific behavioral and attitudinal factors rather than by general demographic or ideological characteristics. The strongest predictors in the model (sociosexual behavior, moral beliefs about infidelity, and relationship com-

mitment) all directly relate to sexual behavior, relational values, and partner investment. This pattern directly supports the findings of Haseli et al. (2021), who argue that “incompatibility of interpersonal characteristics is more likely associated with infidelity than incompatibility of intrapersonal characteristics.”

B. XGBoost

For the XGBoost model, We created a SHAP summary plot to analyze the features’ significances at a deeper level.

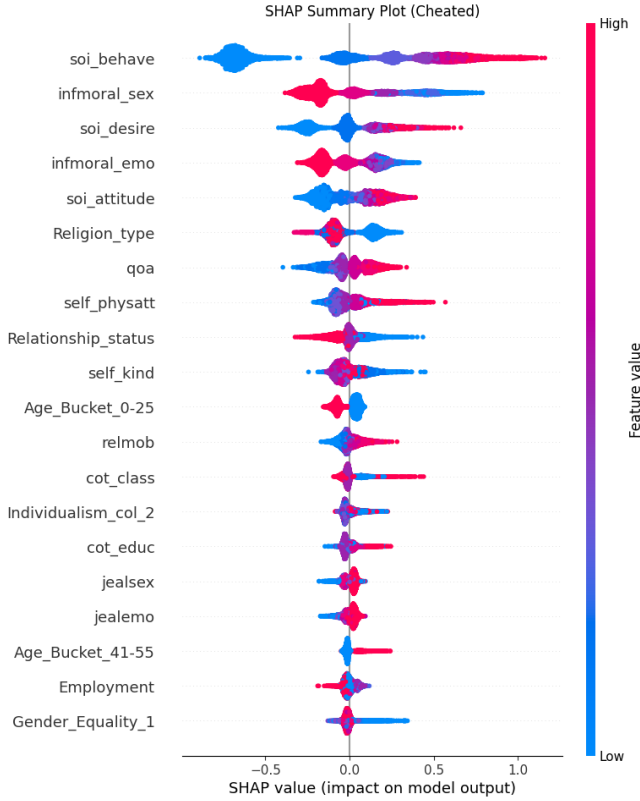


FIG. 5. XGBoost SHAP Summary Plot

Based on the SHAP plot, there is a lot of confusion in the variables (FIG 5). However, it can be seen that the more casual sexual partners a participant has had, the more of a positive impact on a model there is, but if they have had a very small amount of casual partners, then the model is more likely to have listed them as not cheated. This feature has the largest effect on the model, almost double all the other variables. A similar spread, though not as expansive, is also visible in the participant’s desire and attitude to casual sex. Though there is some confusion, it can be seen that those with less education have a negative impact, while those with higher education have a larger positive impact on the model. Also noticeable is that if a person believes they are not very kind, there is a positive effect on the model. As for gender equality, those who believe women belong in the kitchen are more likely to be categorized as cheaters by the model.

Next, looking at the partial difference plots for the top features, more granular patterns of a participant’s likelihood of cheating in the XGBoost model can be analyzed. The most significant feature was soi_behave, which is the amount of casual sexual partners the person has had.

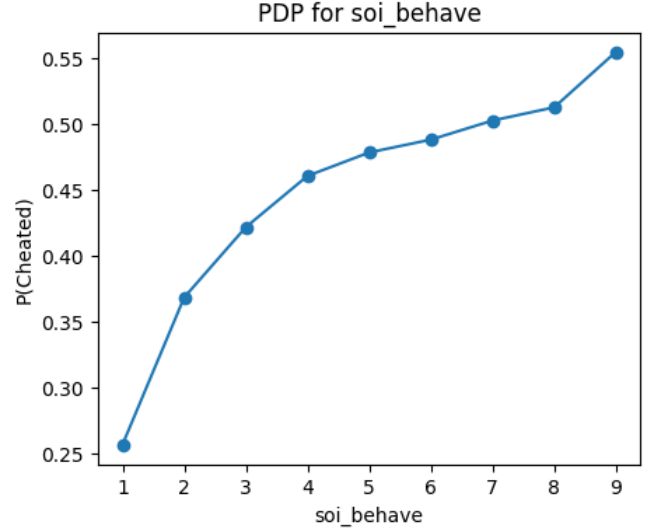


FIG. 6. PDP Plot: soi_behave

Based on the PDP plot above, it can be seen that a person is more likely to be predicted to cheat as the number of partners increase (FIG 6). People who have had more casual sexual experiences may be more used to being in physical relationships without the added responsibility of an emotional relationship, and thus, more willing to look for partners outside of their current relationship. A person’s desire and attitude to casual sex reflects the PDP graph for behavior, also increasing with a more positive attitude and greater desire for casual relationships.

The second most significant feature was infmoral_sex, which is the participant’s opinion of sexual infidelity, listing from a scale of 1 (not at all) to 7 (extremely) of how immoral they think it is.

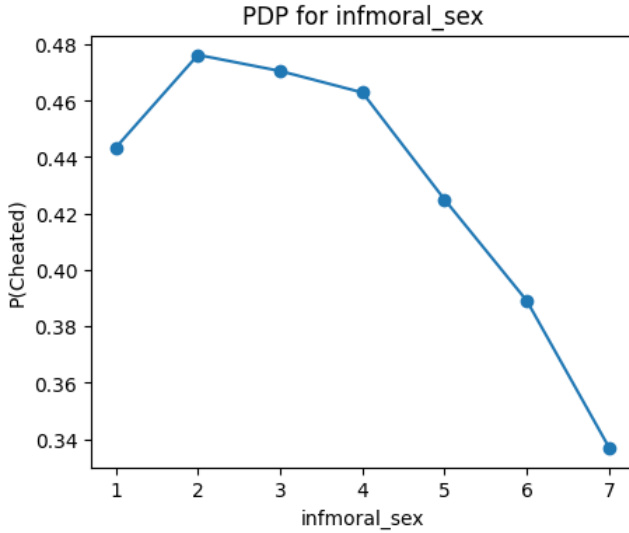


FIG. 7. PDP Plot: infmoral_sex

As seen in the PDP graph above, those who think it is very reprehensible tend to not cheat (FIG 7). However, rather than the max occurring to those who think it is not at all bad, it is centered around 2 - 4 immoral severity. This may be due to people in those ranges who only believe it to be slightly immoral might find a taboo appeal to cheating. How immoral they believe emotional infidelity also decreases their likelihood for cheating, much like sexual infidelity; however its PDP is much smoother, consistently decreasing from 1 to 7.

The third most significant feature was Religion_type, which is the participant's religion.

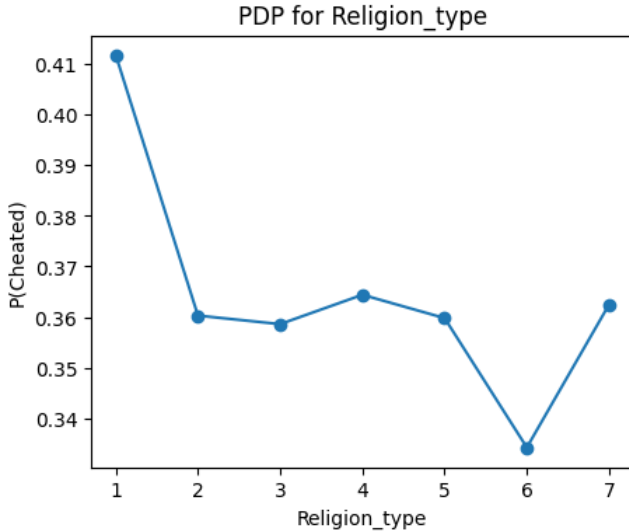


FIG. 8. PDP Plot: Religion_type

The PDP graph describes how those who are Christian (1) are the most likely to cheat out of any religion type and Jewish (6) participants are the least likely, while all the rest of the religion types (2-5), or those with no religion (7), have the

same low effect on the model (FIG 8).

With the feature importance analysis, it can be seen that both the logistic regression and XGBoost models listed soi_behave (casual sexual partners count) as one of the most important variables in determining whether someone is cheating. They both also listed the attitudes and desire toward casual sex, the participant's moral view toward emotional and sexual infidelity, and their confidence on finding a new partner as impactful on the model predictions. Thus, as both models noticed these features as being important, we can make the assumption that these are the most important factors overall in determining someone's infidelity status.

VII. DISCUSSION

Romantic relationships have been the focus of extensive research, with infidelity receiving particular attention due to its role in the breakdown of many partnerships. According to a study by Haseli et al., "incompatibility of interpersonal characteristics is more likely associated with infidelity than incompatibility of intrapersonal characteristics" (2021). This aligns with our findings that relational behavior variables like causal sexual tendencies are solid predictors of infidelity, whereas more generic traits like gender-equality attitudes and employment status contributed minimally.

However, identifying what constitutes infidelity is difficult, as studies show that definitions vary widely across individuals and cultures. These variations shape not only which behaviors people report but also how strongly different traits predict infidelity. For example, a cross-cultural study led by UCLA Health found that in societies where men are not expected to be heavily involved in daily childcare, people tend to be less prone to jealousy (2019). Similarly, in cultures where multiple sexual relationships are considered normal, jealousy is often muted. These findings highlight the extent to which cultural norms influence emotional responses and relationship expectations, which in turn affect both self-reported infidelity and the predictive power of traits such as jealousy or commitment.

Our findings contribute to existing literature by demonstrating that some predictors of infidelity, such as sociosexuality and morality, are consistent across cultures. Meanwhile, other traits' predictive power depends on the cultural context. Existing research about romantic relationships has primarily focused on western, educated, industrial, rich, and democratic countries limiting the ability to distinguish universal predictors from culturally contingent ones. By training our machine learning models on a culturally-diverse and large dataset, our results can highlight patterns that are not prominent in homogenous samples. For example, the relatively weak predictive power of demographic variables and self-identified traits support that interpersonal traits do not strongly influence infidel behaviors. Instead, our results suggest that behavioral and attitudinal factors are more fundamental drivers of infidelity across different cultures.

Beyond theoretical relevance, understanding predictors of infidelity have practical meaning for understanding how cultural norms influence relationship behaviors. For example, adultery in South Korea is a illegal under the Korean Civil Act

and compensation for mental damages can be claimed (Haeon Divorce). In such contexts, social and legal consequences raise the stakes of infidelity, which may influence both how people behave and how they report behavior. Our findings regarding infidelity predictors imply that interventions aiming to reduce infidelity would be more effective if they focus on understanding and reshaping attitudes and behaviors instead of broad demographic or ideological factors. Moreover, cross-cultural predictive models like ours can inform policymakers or support services about which psychological and relational factors tend to be more relevant in different cultural contexts, potentially guiding culturally sensitive approaches to relationship counseling, education, or legal reform. However, given the personal and sensitive nature of infidelity, applying predictive models in real-world contexts raises concerns about privacy, stereotyping, and potential harm. Thus, any practical use of these findings should be guided by strict ethical standards and cultural sensitivity.

Despite the insights derived from our models, there are several limitations within our dataset. First, our outcome variable relies on self-reported infidelity, which is subject to social desirability bias and may vary in accuracy across cultures that differ in attitudes toward cheating. Additionally, because the dataset is cross-sectional, our models can identify associations but cannot determine causal relationships or temporal patterns leading up to infidelity. Furthermore, our selected features cannot capture all factors relevant to infidelity, such as attachment styles, relational histories, or contextual life stressors. Future research could address these limitations by incorporating longitudinal or behavioral data, and developing models within individual cultural groups to compare with cross-cultural models to better understand how specific traits interact. Expanding the feature set to include communication patterns, relationship duration, or digital behavior could also sharpen predictive power. This would deepen our understanding of both universal and culture-specific pathways to infidelity and enhance the practical value of computational models in relationship research.

VIII. CONCLUSION

This study illustrates how machine learning can be used as a complementary tool in relationship research to quantify behavioral patterns that are difficult to capture through traditional analytical approaches. By combining interpretable and flexible modeling techniques with a large, cross-cultural dataset, we show that infidelity can be predicted with reasonable accuracy and calibration using a limited set of psychologically meaningful features. Importantly, the consistency of key predictors across models suggests that certain behavioral tendencies, particularly those tied to sexual norms and relational decision-making, form a stable statistical foundation for understanding infidelity at scale.

Beyond model performance, this work highlights the value of prioritizing calibration and interpretability when addressing socially sensitive outcomes. Well-calibrated probability estimates allow future predictions to be framed as likelihoods rather than deterministic labels, reducing the risk of overconfidence

or misinterpretation. The comparison between logistic regression and XGBoost further demonstrates that increased model complexity does not necessarily yield proportionate gains, reinforcing the practical relevance of simpler, more transparent approaches in applied social science contexts.

Ultimately, this research underscores the potential for data-driven methods to inform broader discussions about relationship dynamics while maintaining ethical caution. Rather than serving as diagnostic tools for individual behavior, models are best suited for population-level insight and hypothesis generation. As computational methods continue to intersect with the study of human relationships, future work should emphasize responsible deployment, cultural sensitivity, and methodological transparency to ensure that predictive analytics enhance our understanding of complex social phenomena.

IX. ACKNOWLEDGMENTS

We would like to thank Dr. Arya Farahi at The University of Texas at Austin for his support and guidance throughout our research.

X. AUTHOR DECLARATIONS

A. Conflict of Interest Statement

The authors have no conflicts to disclose.

B. AI Usage

We have used AI to assist with code debugging and grammatical error fixes in our presentations and this report. While we verified our understanding of the metrics used for the models, we intentionally did not use AI for interpretation or analysis as we wanted to ensure that we were considering the practical implications of the results from our analysis as quantitative outputs cannot capture the full complexities of data, especially that which relates to human behavior.

C. Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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